

Vol 10 Chapter 8 — Representation Theory

Keeper (author pass — deep math/physics revision)

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Chapter 8 — Representation Theory

Level 1 — one sentence

Representation theory expresses abstract groups as concrete linear actions on vector spaces, with irreducible representations the building blocks — and the BST substrate’s Wallach 1976 unitary representations of $SO_0(p, 2)$ supported on bounded symmetric domains organize the substrate’s K-type spectrum that anchors particle masses (Vol 2 Ch 6) and the substrate operator zoo (Vol 5).

Level 2 — graduate-physicist precision

8.1 Representations

A representation $\rho : G \rightarrow GL(V)$ of group G on vector space V is a homomorphism: $\rho(gh) = \rho(g)\rho(h)$.

Irreducible: no proper invariant subspaces. Completely reducible: every rep is a direct sum of irreducibles (true for compact, semisimple, finite groups).

Character $\chi_\rho(g) = \text{Tr } \rho(g)$: invariant under conjugation; complete invariant for rep.

8.2 Compact Lie group representation theory

For compact Lie group K : every irreducible representation is finite-dimensional and unitarizable. Classified by highest weights (Cartan-Weyl theory).

Weyl character formula: χ_λ in terms of highest weight λ and Weyl group.

8.3 $SO(5)$ and $SO(5) \times SO(2)$ representations

For $SO(5)$ (rank 2): irreducible reps labeled by highest weights (λ_1, λ_2) with $\lambda_1 \geq \lambda_2 \geq 0$ integers (or half-integers for spinor reps).

Dimensions via Weyl formula. Casimir eigenvalues: $C_2(\lambda) = \lambda_1(\lambda_1 + 3) + \lambda_2(\lambda_2 + 1)$.

The (1,1) rep has Casimir $= 4 + 2 = 6 = \text{BST primary } C_2$. The (1,0) “vector” has $C_2 = 4$. The (1/2, 1/2) “spinor” has $C_2 = 15/4 + 3/4 = 9/2$.

For $SO(5) \times SO(2)$: pairs (SO(5) highest weight) $\times$ (SO(2) weight k). K-types of BST.

8.4 Wallach representations

Wallach 1976: constructed family of unitary representations of $SO_0(p, 2)$ supported on bounded symmetric domains. Each is realized on a Bergman-type space.

For BST $SO_0(5, 2)$: Wallach family organizes into 5 layers (corresponding to BST primary $n_C = 5$). Used to anchor particle mass hierarchy via Wallach layer index (Vol 2 Ch 6).

8.5 Applications

- Standard Model particles classified by representations of $SU(3) \times SU(2) \times U(1)$
- Atomic spectroscopy: $SO(3)$ angular momentum reps give multiplet structure
- Nuclear shell model: $SU(3)$ Elliott model
- Condensed matter: lattice group reps classify band structure
- BST: $SO_0(5, 2)$ Wallach reps anchor everything

8.6 K-audit anchors

- Wallach 1976 (L1 ESTABLISHED): unitary reps on bounded symmetric domains
- Vol 5 Ch 1: K-type decomposition of substrate Hilbert space
- Vol 2 Ch 6: mass hierarchy from Wallach layer index

Level 3 — 5th-grader accessibility

Representation theory: how abstract groups act on vector spaces. Irreducible reps = basic building blocks. For Lie groups: classified by highest weights (Cartan-Weyl). For $SO(5)$: irreducible reps labeled by (λ_1, λ_2) . The (1,1) rep has Casimir = 6 = BST primary C_2 . Wallach 1976: classified unitary reps of $SO_0(p, 2)$ on bounded symmetric domains. BST uses Wallach's classification to organize the substrate's K-type spectrum. Particle masses follow Wallach layer index (Vol 2 Ch 6).

What comes next

Chapter 9 develops special functions.

Where to look this up

- Fulton and Harris, *Representation Theory*
- Wallach 1976
- BST: Vol 5 Ch 1; Vol 2 Ch 6